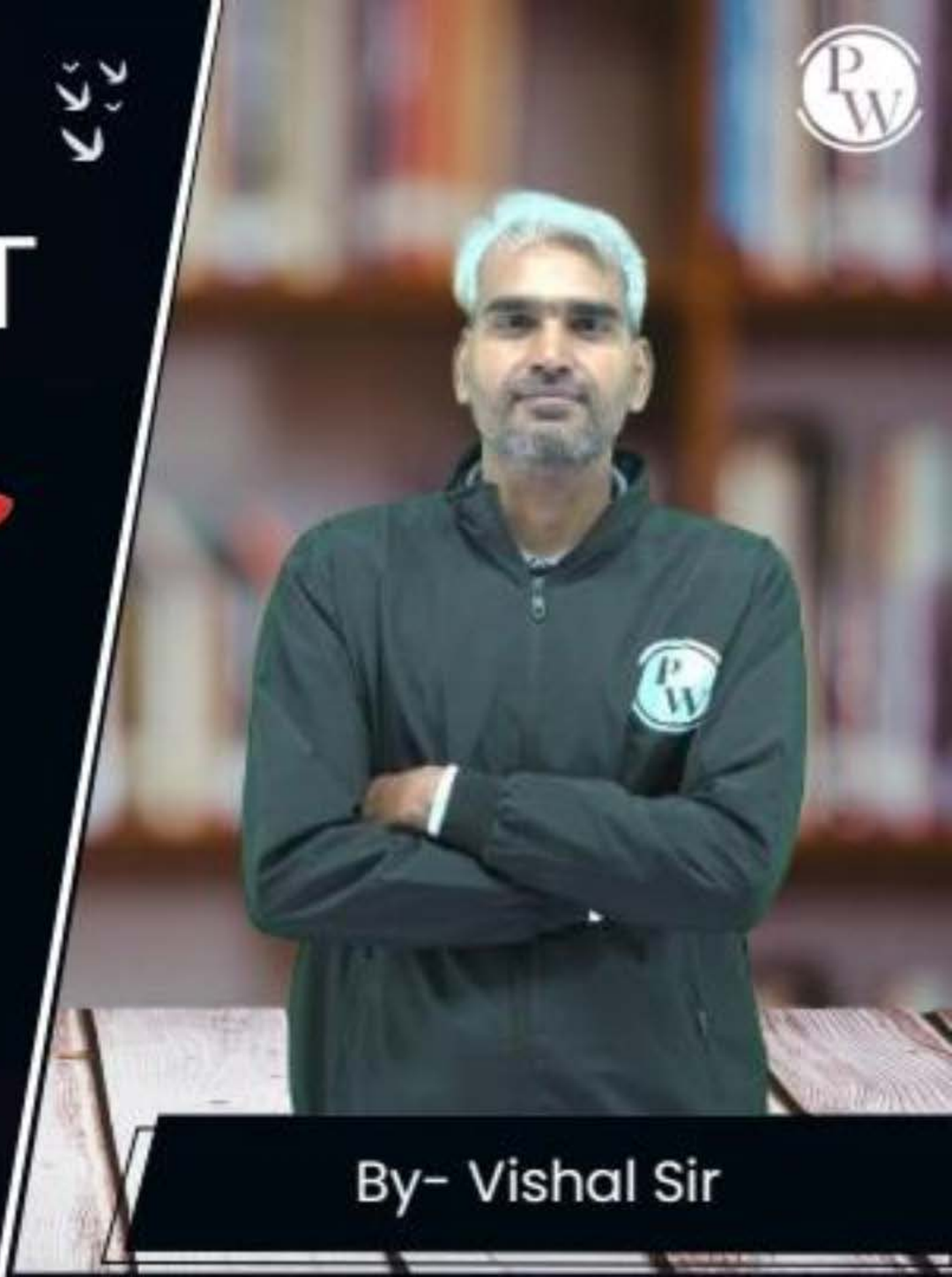
Computer Science & IT



Database Management System

Relational Model & Normal Forms

Lecture No. 16



By- Vishal Sir

Recap of Previous Lecture



- **Topic** Identifying the normal form of relation
- **Topic** Decomposition of relation up to BCNF

Slide

Topics to be Covered



Topic — Decomposition of relation up to BCNF

Topic — Important properties

Slide

Possible types of non-trivial FDs which may cause redundancy in the relation



Type ① Proper subset of a C.K → Non-prime attributes ✓

Type ② (Proper Subset of a Candidate key + Non-prime attributes) → Non-prime attribute

Type ③ Non-prime attributes → Non-prime attributes

Non-Prime Attributes → P.S.C.K. {Such FDs are not Possible}

Proper subset of a C.K → Proper subset of same C.K Such FD is not Possible

Type ④ Proper subset of one C.K → Proper Subset of some other C.K

Type ⑤ (Proper Subset of one Candidate key + Non-prime attributes) → Proper Subset of some other C.K

#e.g. Given R(ABCDE) and F={AB→C, C→D, B→E }

Find the normal form of the relation, and if relation is not already in BCNF, then decompose the relation up to BCNF.

C.K = (AB)

P.A = {A, B}

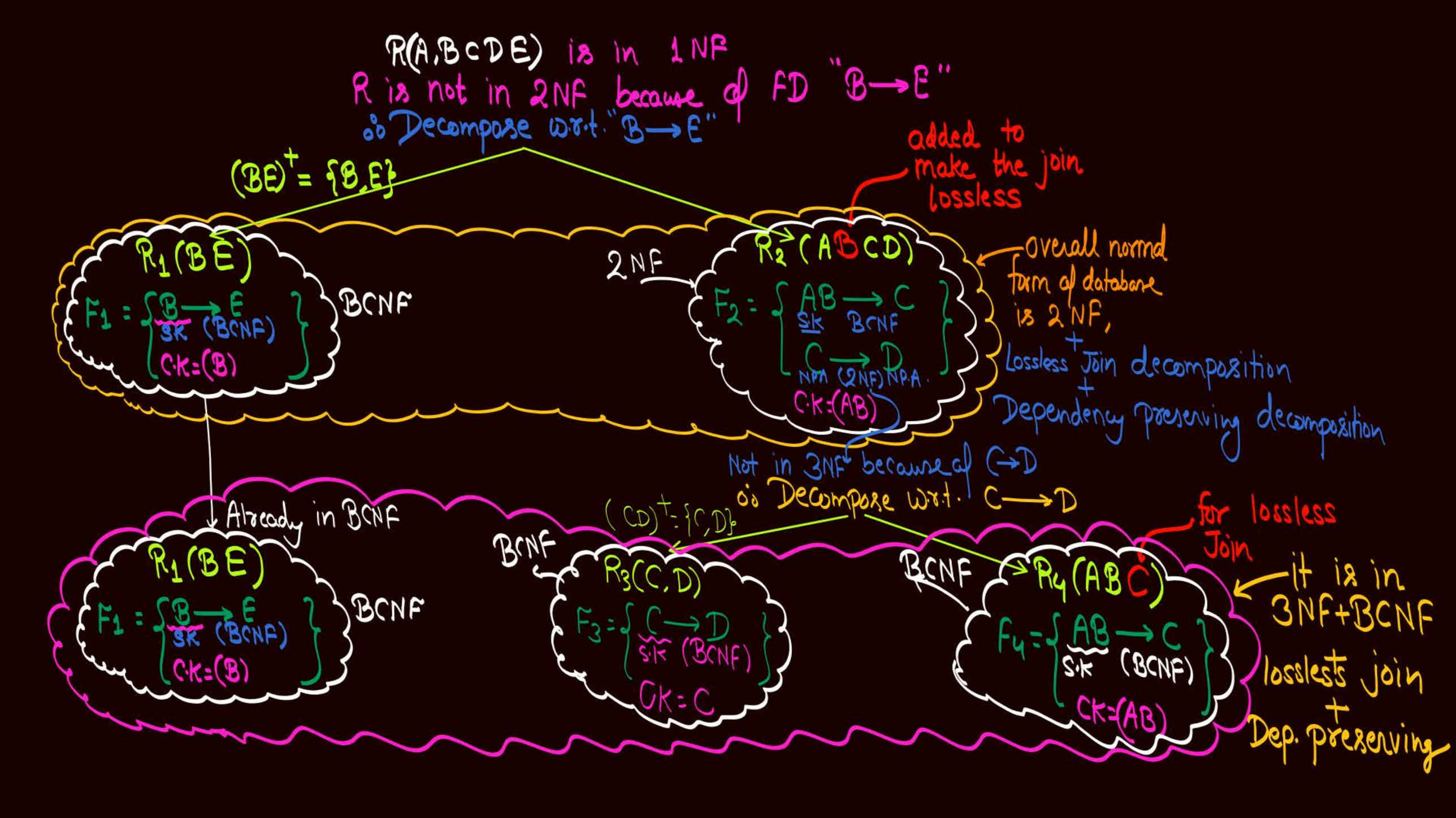
N.P.A = {C, D, E}

F = { AB → C (S.K) (BCNF)

C → D (NP.A → NP.A) (2NF)

B → E (PSCK → N.P.A) (1NF) }

Normal form of Relation is 1NF


R(A,B,C,D,E) is in 1NF
R is not in 2NF because of FD "B → E"
∴ Decompose w.r.t. "B → E"
(BE)+ = {B, E}
added to make the join lossless
R1(B E)
F1 = { B → E (BCNF) SK ; C.K = (B) }
BCNF
2NF
R2(A B C D)
F2 = { AB → C SK BCNF ; C → D NPA (2NF) N.P.A. ; C.K = (AB) }
overall normal form of database is 2NF,
+ Lossless Join decomposition
+ Dependency preserving decomposition
Not in 3NF because of C → D
∴ Decompose w.r.t. C → D
(CD)+ = {C, D}
Already in BCNF
R1(B E)
F1 = { B → E (BCNF) SK ; C.K = (B) }
BCNF
BCNF
R3(C, D)
F3 = { C → D (BCNF) S.K ; CK = C }
BCNF
R4(A B C)
F4 = { AB → C (BCNF) S.K ; CK = (AB) }
for lossless Join
it is in 3NF+BCNF
lossless join
+
Dep. preserving

#e.g. Given R(ABCDEF) and F={A→BCDEF, BC→ADEF, D→E , E→F}

Find the normal form of the relation, and if relation is not already in BCNF then decompose the relation up to BCNF.

C.K = {A}, {BC}

P.A = {A, B, C}

N.P.A = {D, E, F}

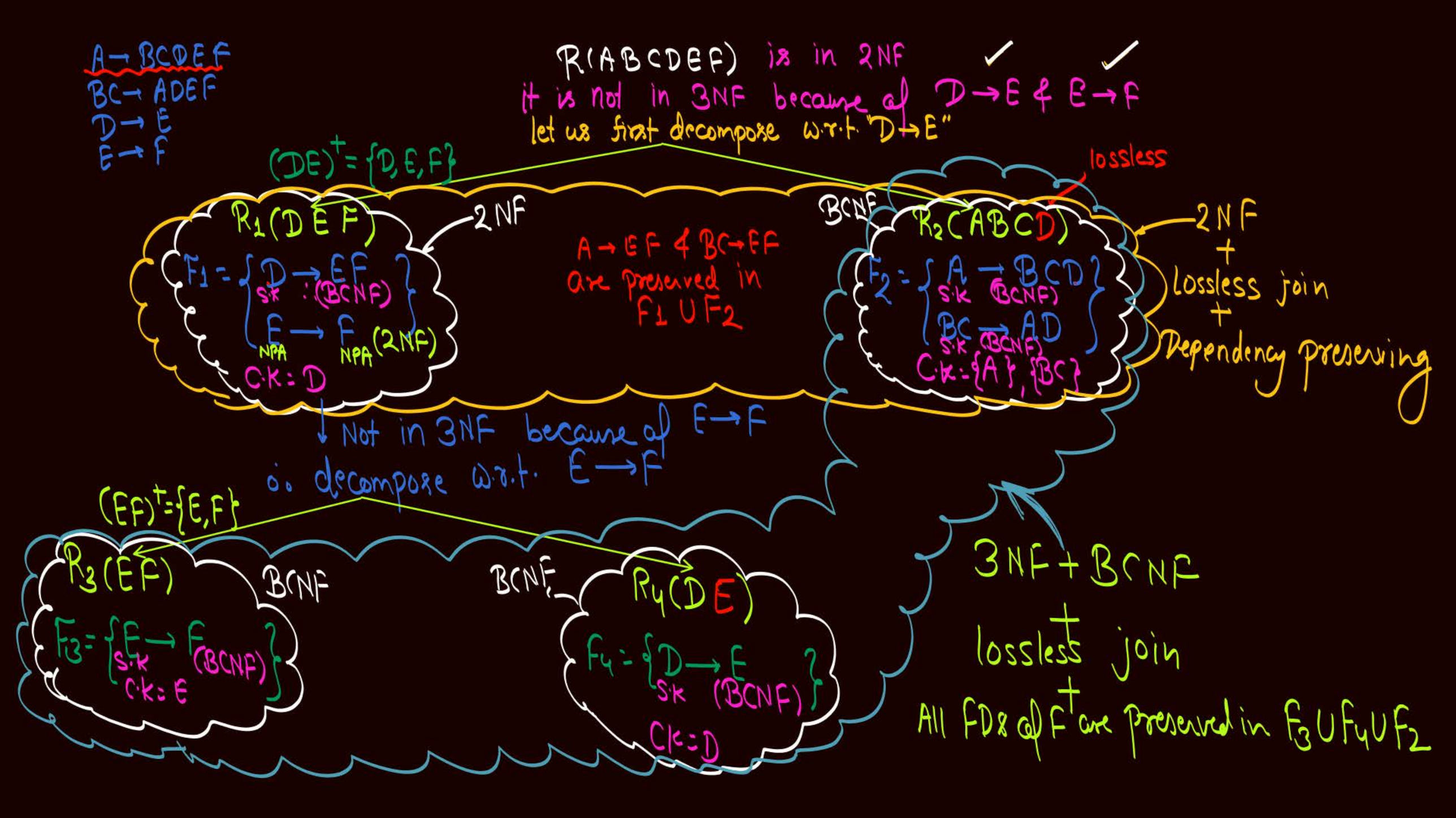
A → BCDEF
S.K ∴ (BCNF)

BC → ADEF
S.K ∴ (BCNF)

D → E
NPA (2NF) NPA.

E → F
NPA (2NF) NPA

Normal Form of Relation is "2NF"


A → BCDEF
BC → ADEF
D → E
E → F
R(ABCDEF) is in 2NF
it is not in 3NF because of D → E & E → F
let us first decompose w.r.t. "D → E"
(DE)+ = {D, E, F}
R1(DEF)
F1 = { D → EF (S.K : BCNF), E → F (NPA NPA (2NF)) }
C.K : D
2NF
A → EF & BC → FF are preserved in F1 ∪ F2
BCNF
R2(ABCD)
F2 = { A → BCD (S.K (BCNF)), BC → AD (S.K (BCNF)) }
C.K :- {A}, {BC}
lossless
2NF + lossless join + Dependency preserving
Not in 3NF because of E → F
i. decompose w.r.t. E → F
(EF)+ = {E, F}
R3(EF)
F3 = { E → F (S.K (BCNF)) }
C.K : E
BCNF
BCNF
R4(DE)
F4 = { D → E (S.K (BCNF)) }
C.K :- D
3NF + BCNF + lossless join + All FDs of F+ are preserved in F3 ∪ F4 ∪ F2

H.W. Try the decomposition upto BCNF, while the first decomposition is w.r.t. "E → F"

R(ABCDEF) is in 2NF

it is not in 3NF because of D → E & E → F

let us first decompose w.r.t "E → F"

$(EF)^+ = \{EF\}$

$R_1(EF)$

$F_1 = \{E \to F\}$ (BCNF)

CK: E

BCNF

2NF + lossless + Dep. preserving

2NF

$R_2(ABCDE)$

$F_2 = \{A \to BCDE$ (BCNF$)$, $BC \to ADE$ (BCNF), $D \to E$ (2NF$)\}$

CK: (A), (BC)

$(DF)^+ = \{D, E, F\}$

$R_3(DE)$

$F_3 = \{D \to E$ (BCNF$)\}$

CK: D

$R_4(ABCD)$

$F_4 = \{A \to BCD$, $BC \to AD\}$ BCNF, BCNF

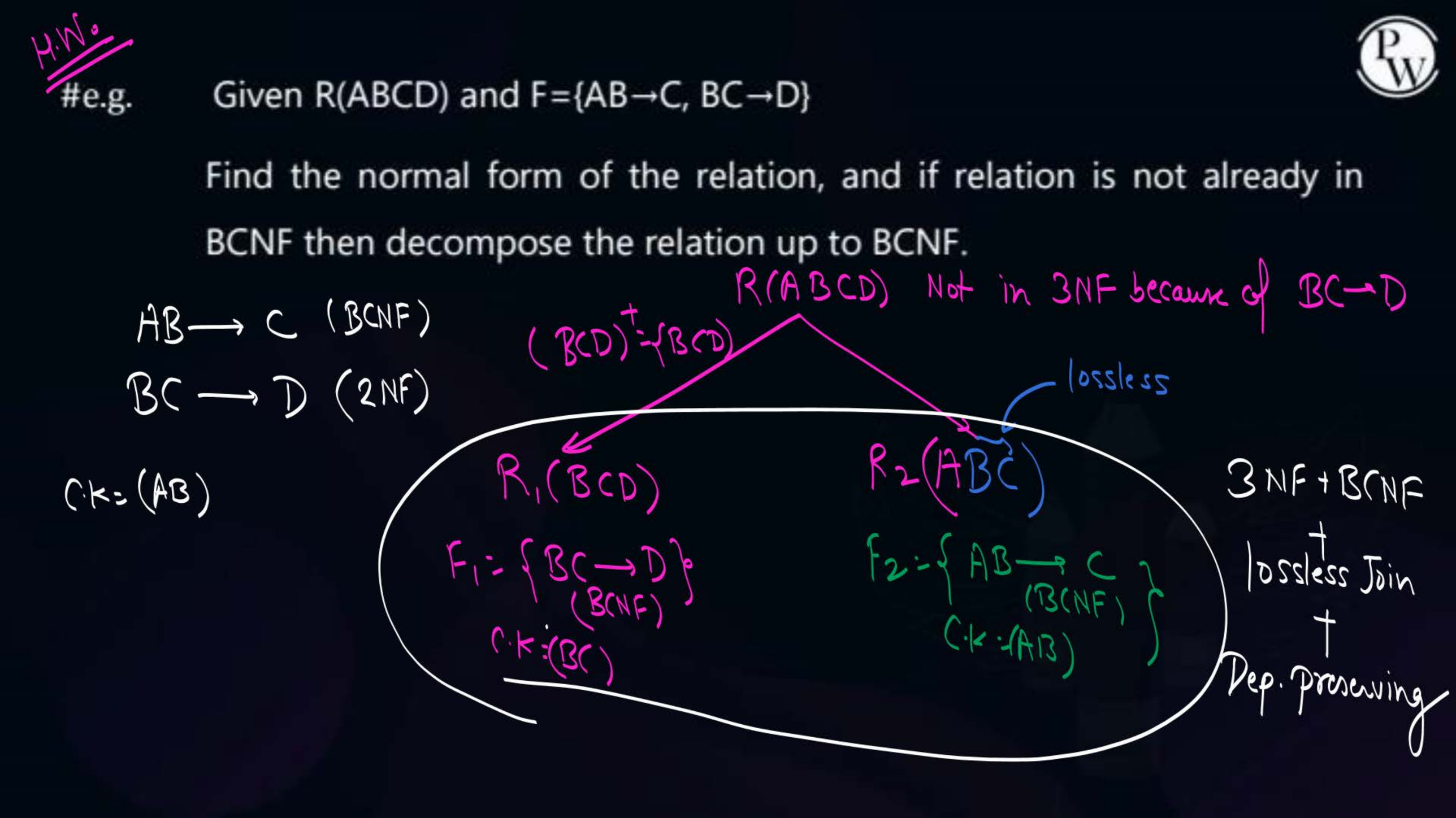
CK: (A), (BC)

{ 3NF + BCNF + lossless + Dep. preserving

H.W.

#e.g. Given R(ABCD) and F={AB→C, BC→D}

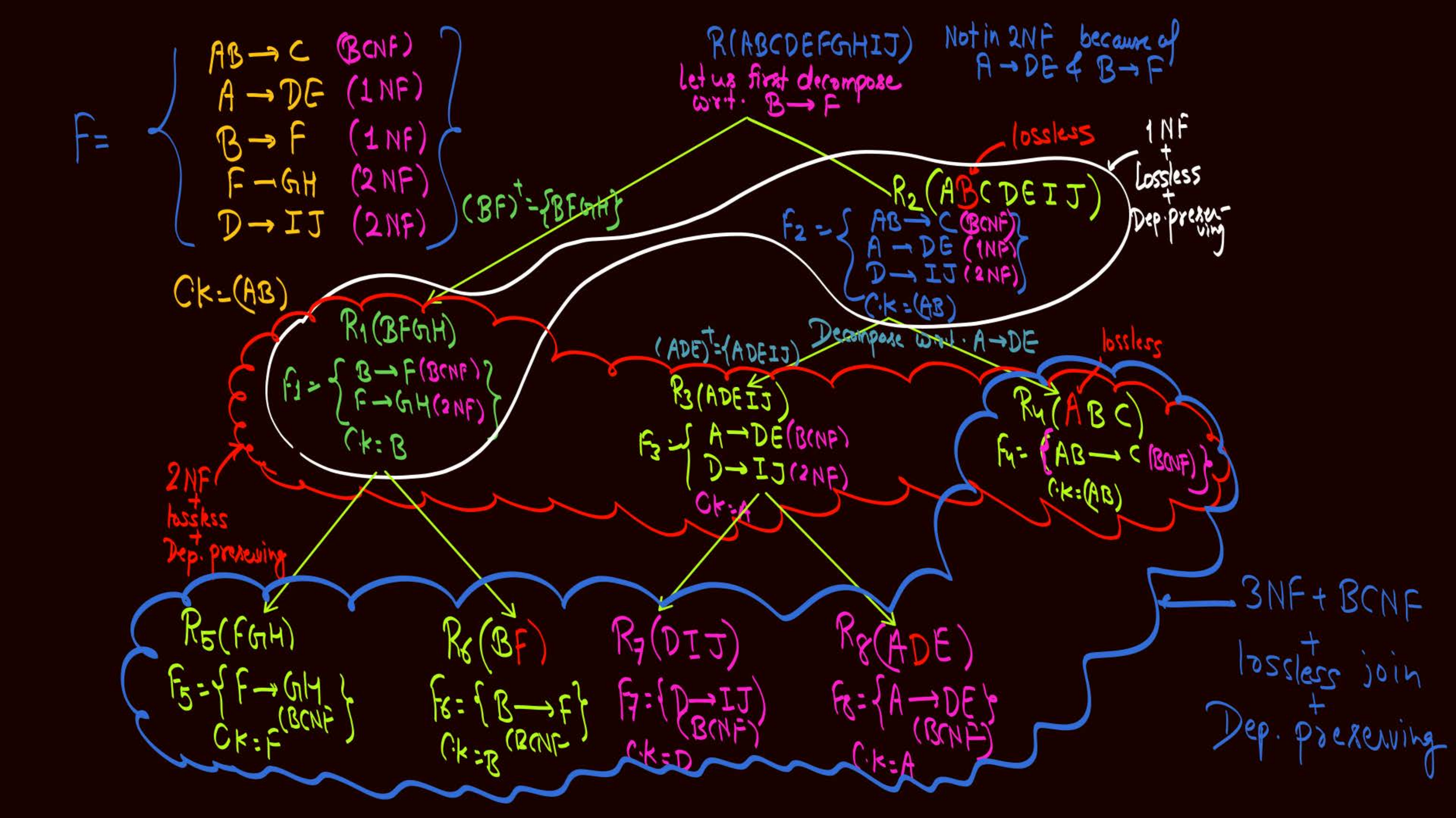
Find the normal form of the relation, and if relation is not already in BCNF then decompose the relation up to BCNF.



H.W.

#e.g. Given R(ABCDEFGHIJ) and F={AB→C, A→DE, B→F, F→GH, D→IJ}

Find the normal form of the relation, and if relation is not already in BCNF then decompose the relation up to BCNF.


F= AB → C (BCNF)
A → DE (1NF)
B → F (1NF)
F → GH (2NF)
D → IJ (2NF)
R(ABCDEFGHIJ)
Not in 2NF because of A → DE & B → F
Let us first decompose w.r.t. B → F
(BF)+ = {BFGH}
lossless
1NF + Lossless + Dep. preserving
R2(ABCDEIJ)
F2 = { AB → C (BCNF), A → DE (1NF), D → IJ (2NF) }
CK = (AB)
R1(BFGH)
F1 = { B → F (BCNF), F → GH (2NF) }
CK: B
(ADE)+ = {ADEIJ}
Decompose w.r.t. A → DE
lossless
R3(ADEIJ)
F3 = { A → DE (BCNF), D → IJ (2NF) }
CK: A
R4(ABC)
F4 = { AB → C (BCNF) }
CK: (AB)
2NF + lossless + Dep. preserving
R5(FGH)
F5 = { F → GH (BCNF) }
CK: F
R6(BF)
F6 = { B → F } (BCNF)
CK: B
R7(DIJ)
F7 = { D → IJ } (BCNF)
CK: D
R8(ADE)
F8 = { A → DE } (BCNF)
CK: A
3NF + BCNF + lossless join + Dep. preserving

H.W.

#e.g. Given R(ABDLPT) and F={B→PT, T→L, A→D}

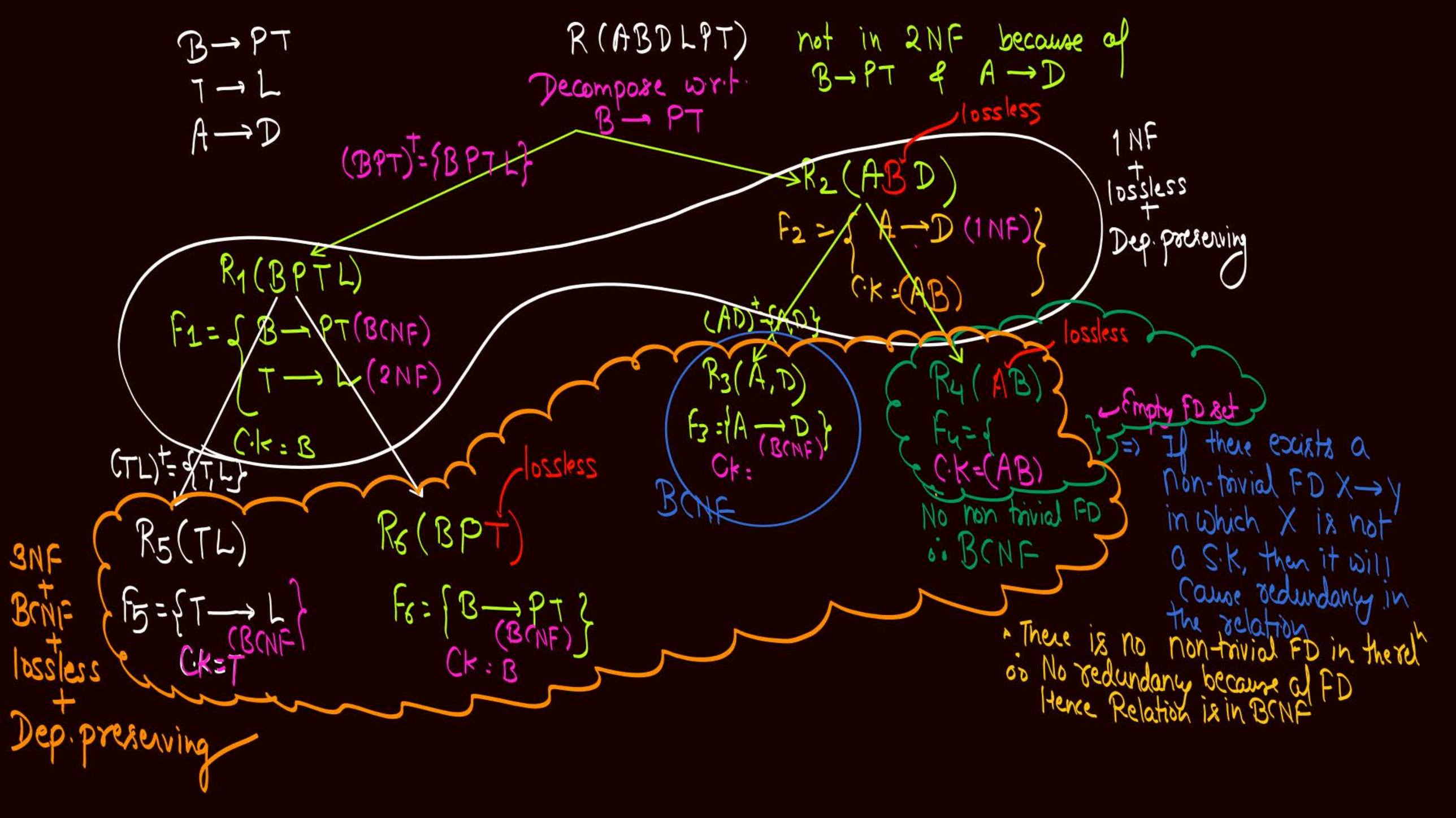
Find the normal form of the relation, and if relation is not already in BCNF then decompose the relation up to BCNF.

$B \longrightarrow PT$ (1NF)

$T \longrightarrow L$ (2 NF)

$A \longrightarrow D$ (1 NF)

C.K = (AB)


B → PT
T → L
A → D
R(ABDLPT) not in 2NF because of B → PT & A → D
Decompose w.r.t. B → PT
(BPT)+ = {B P T L}
R1(BPTL)
F1 = { B → PT (BCNF), T → L (2NF) }
C.K : B
(TL)+ = {T, L}
R2(ABD)
F2 = { A → D (1NF) }
C.K : (AB)
lossless
1NF + lossless + Dep. preserving
(AD)+ = {A D}
R3(A, D)
F3 = { A → D (BCNF) }
Ck :
BCNF
R4(AB)
F4 = { }
C.K = (AB)
No non trivial FD
∴ BCNF
lossless
Empty FD set
=> If there exists a non-trivial FD X → y in which X is not a S.K, then it will cause redundancy in the relation
∴ There is no non-trivial FD in the rel^n
∴ No redundancy because of FD
Hence Relation is in BCNF
R5(TL)
F5 = { T → L (BCNF) }
C.K = T
R6(BPT)
F6 = { B → PT (BCNF) }
Ck : B
lossless
3NF + BCNF + lossless + Dep. preserving

Note:- If there is no non-trivial FD in a relation, then Candidate key of that relation will be formed by combining all the attributes of that relation, and such relation will always be in BCNF

#e.g. Given R(ABCDE) and F:{A→BC, CD→E, B→D, E→A}

Find the normal form of the relation, and if relation is not already in BCNF then decompose the relation up to BCNF.

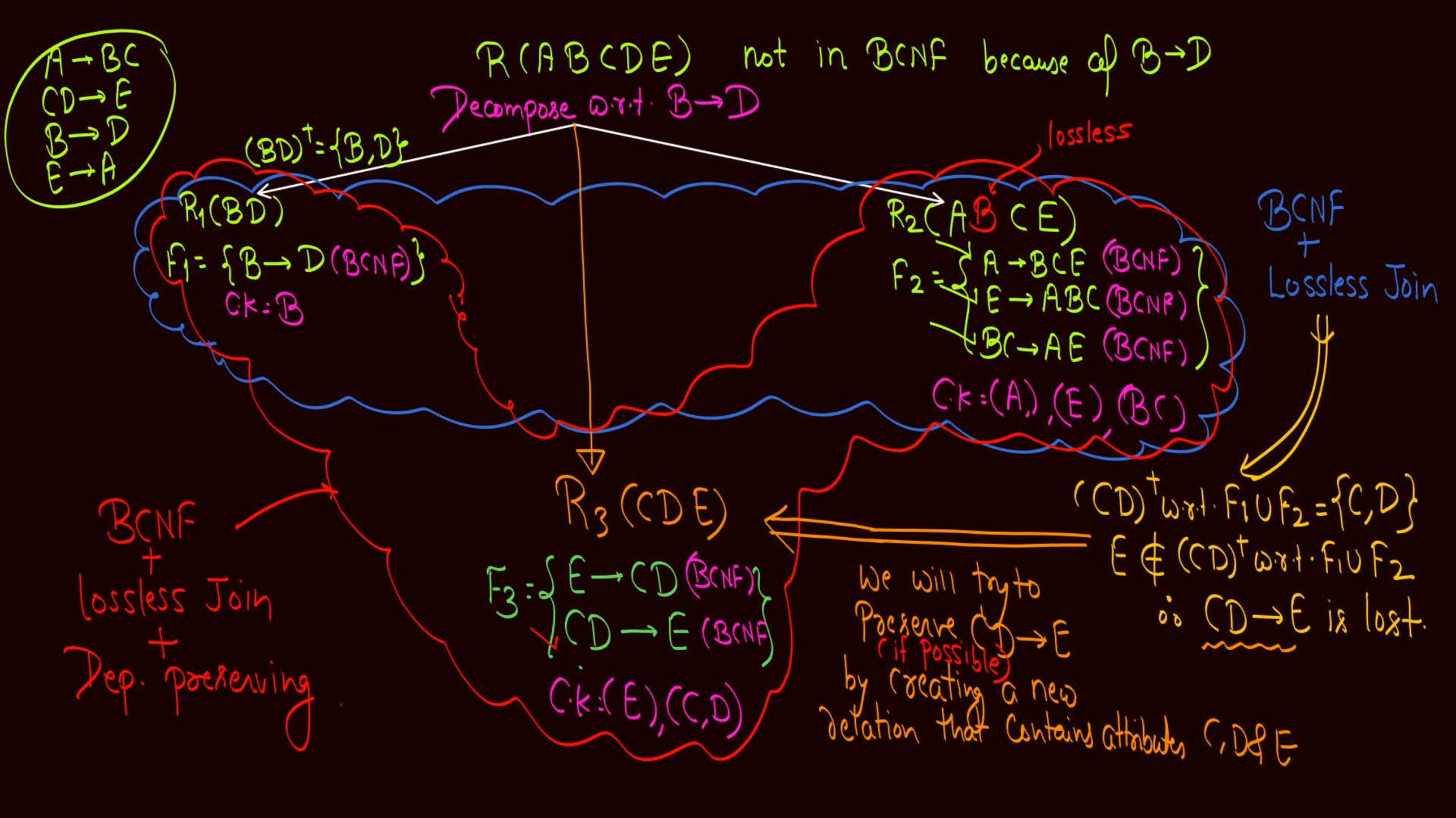
A → BC (BCNF)
S.K

CD → E (BCNF)
S.K

B → D (3NF)
P.S.C.K P.S.C.K

E → A (BCNF)
S.K

C.K = (A), (E), (CD), (CB)


A → BC
CD → E
B → D
E → A
R(ABCDE) not in BCNF because of B → D
Decompose w.r.t. B → D
(BD)⁺ = {B, D}
lossless
R1(BD)
F1 = {B → D (BCNF)}
Ck : B
R2(ABCE)
F2 = {A → BCE (BCNF), E → ABC (BCNF), BC → AE (BCNF)}
Ck : (A), (E), (BC)
BCNF + Lossless Join
R3(CDE)
F3 = {E → CD (BCNF), CD → E (BCNF)}
Ck : (E), (C, D)
BCNF + lossless Join + Dep. preserving
(CD)⁺ w.r.t. F1 ∪ F2 = {C, D}
E ∉ (CD)⁺ w.r.t. F1 ∪ F2
∴ CD → E is lost.
We will try to Preserve CD → E (if Possible) by creating a new relation that Contains attributes C, D & E

#e.g. Given R(ABCD) and F={AB→CD, D→A }

Find the normal form of the relation, and if relation is not already in BCNF then decompose the relation up to BCNF.

AB → CD
S.K (BCNF)

C.k. = (AB), (DB)

D → A (3NF)
P.S. of one C.k
P.S. of another C.k

$AB \rightarrow CD$
$D \rightarrow A$

$R(ABCD)$ is not in BCNF because of $D \rightarrow A$

∴ Decompose w.r.t. $D \rightarrow A$

$(AD)^+ = \{A, D\}$

$R_1(AD)$

$F_1 = \{ D \rightarrow A \}$ (BCNF)

C.K = (D)

$R_2(BCD)$ — lossless

$F_2 = \{ BD \rightarrow C$ s.t. (BCNF), C.K = (BD) $\}$

BCNF + lossless +

$(AB)^+$ w.r.t. $F_1 \cup F_2 = \{A, B\}$

$CD \notin (AB)^+$ w.r.t. $F_1 \cup F_2$

∴ $AB \rightarrow CD$ is lost

To preserved $AB \rightarrow CD$, if we create a new sub-relation $R'(ABCD)$, then $D \rightarrow A$ will be a member of R' as well and $D \rightarrow A$ will remain in 3NF

↳ i.e. We will not be able to preserve $AB \rightarrow CD$ if we want to improve the normal form of database to BCNF

Note:- (1) Upto 3 NF we can always ensure lossless join decomposition as well as dependency Preserving decomposition

(2) While decomposing a relation into BCNF some times (Not always) it may not be possible to preserve some of the functional dependencies of the original relation, but we can always ensure lossless join decomposition.

(3) Most adequate normal form of the database is 3NF, because we can always ensure lossless join decomposition as well as dep. preserving decomposition {i.e. No loss of information}

④ In a relation if all attributes are prime attributes, then the relation is at least in 3NF.

⑤ If all candidate keys of the relation are simple Candidate keys, then partial dependency can not exist in the relation, therefore relation will be at least in 2NF.

⑥ In a relation if all attributes are prime attributes, and all candidate keys of the relation are simple Candidate keys, then relation is always in BCNF

⑦ A binary relation {i.e. a relation with exactly two attributes} is always in BCNF.

Let R(AB) is a relation with two attributes, then only four different types of FD set are possible





2 mins Summary

Topic Decomposition of relation up to BCNF

Topic Important properties

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t.me/vishalRawtiyapw

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Capital

THANK - YOU

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